

Technology #01

Multi-Segment Flow Control with Bayesian
Occlusion Prediction

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION
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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Multi-Segment Flow Control with Bayesian Occlusion Prediction

WHY DOES IT MATTER?

Catheter obstruction causes 30-50% of shunt failures

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 4 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Hydraulics + Control Systems. 6-step build plan with specific test articles and acceptance criteria. Target buyer: Neuroshunt manufacturer (Medtronic, Integra, Miethke).

WHAT DOES THE BUYER GET?

Multi-segment catheter concept + mechanism description; Bayesian predictor algorithm (225 ablation runs, rate-limiting confirmed); Alpha-distribution controller with graceful degradation protocol; Dual-invariant falsification record (honest failure disclosure); V0 bench prototype BOM (COTS sensors + Arduino); Computational model (svMultiPhysics verified 1D within 16%); This engineering dossier with external precedent

WHAT DOES THE BUYER HAVE TO BUILD?

Production catheter design (dimensioned drawings, tolerances); Implantable-grade flow sensor (COTS → implantable transition); Production controller (Arduino → medical-grade MCU); Manufacturing process for multi-segment catheter; Regulatory submission (PMA Class III); Clinical validation evidence

WHAT IS THE NEXT DECISIVE EXPERIMENT?

V0 bench prototype (4-segment catheter + COTS sensors + Arduino)

WHAT WOULD MAKE US KILL IT?

Multi-segment shows no advantage over single-segment in $\geq 80\%$ of computational scenarios

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 6-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: V0 bench prototype (4-segment catheter + COTS sensors + Arduino)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - V0 bench prototype (4-segment catheter + COTS sensors + Arduino) - 8 weeks

Basis: Build plan WP. Scope: Per-segment flow + ICP under obstruction scenarios. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - Controller with graceful degradation - 8 weeks

Basis: Build plan WP. Scope: Closed-loop performance under dual-invariant conflict. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - Biocompatibility specimens (multi-segment silicone catheter + COTS sensor encapsulation) - 12 weeks (external lab)

Basis: Build plan WP. Scope: ISO 10993 series. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 4 equations. Key assumptions: Multi-segment catheter with N parallel lumens (N=4 in V0 prototype); Per-segment flow sensors provide observable $F_{obs}(t)$; Bayesian predictor can infer occlusion probability from conductance trend; Controller can redistribute flow via α_i adjustment; CSF is Newtonian at shunt flow rates. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: V0 bench prototype (4-segment catheter + COTS sensors + Arduino). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Multi-segment parallel catheter with Bayesian occlusion prediction and alpha-distribution controller

POTENTIAL VALUE DRIVER: Addresses: Catheter obstruction causes 30-50% of shunt failures

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Neuroshunt manufacturer (Medtronic, Integra, Miethke)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Hydraulics + Control Systems

Disciplines: Fluid mechanics, Control engineering, Bayesian statistics, Embedded systems

Architecture: Multi-segment CSF shunt catheter with per-segment flow sensors, Bayesian occlusion predictor, and alpha-distribution controller

Subsystems:

SS-01: Multi-segment catheter - Distribute CSF flow across N segments

SS-02: Per-segment flow sensors - Measure flow rate F_i per segment

SS-03: Bayesian occlusion predictor - Compute $P(\text{occlude}_i | F_{\text{obs}})$ per segment

SS-04: Alpha-distribution controller - Redistribute flow α_i based on $P(\text{occlude}_i)$

2. Mechanism Architecture

Physical Changes: Flow redistribution: alpha values change per-segment flow rates to maintain drainage when one segment occludes

Key Physics: Multi-segment parallel conductance: $G_{\text{total}} = \sum G_i$. When G_j decreases (occlusion), controller increases α_i for surviving segments.

3. Governing Engineering Model

Summary: Multi-segment CSF shunt catheter with Bayesian occlusion prediction and alpha-distribution controller. Flow redistribution across N segments maintains drainage when one segment occludes.

Equations:

$Q = (\pi \cdot r^4 \cdot \Delta P) / (8 \cdot \eta \cdot L)$ [Hagen-Poiseuille per segment]

$G_{\text{total}} = \sum(G_i)$ [parallel conductance across segments]

$P(\text{occlude}_i | F_{\text{obs}})$ updated via Bayesian inference on conductance trend dG_i/dt

$\alpha_i = \text{controller}(P(\text{occlude}_i))$ [flow redistribution policy]

Assumptions:

- Multi-segment catheter with N parallel lumens (N=4 in V0 prototype)
- Per-segment flow sensors provide observable $F_{\text{obs}}(t)$
- Bayesian predictor can infer occlusion probability from conductance trend
- Controller can redistribute flow via α_i adjustment
- CSF is Newtonian at shunt flow rates

Boundary Conditions:

- ICP 5-40 mmHg
- Per-segment flow 0.05-0.5 mL/min
- N segments in parallel

Failure Regimes:

- Multiple simultaneous occlusions (system redundancy exhausted)
- Sensor failure on multiple segments
- Controller instability under rapid pressure changes
- Dual-invariant conflict (peak ICP exceeds 20 mmHg — already FALSIFIED in R332)

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Catheter obstruction causes 30-50% of shunt failures. \$35-50K per revision	VERIFIED SOURCE_DERIVED: HCUP/AHRO
DI-002	Functional requirement	Multi-segment CSF shunt with Bayesian occlusion prediction and pre-emptive occlusion distribution maintaining dual safety	MODELLED
DI-003	Performance acceptance	Multi-segment peak ICP < single-segment in >=80% of scenarios	MODELLED
DI-004	Falsification criterion	Multi-segment shows no advantage over single-segment	MODELLED
DI-005	ICP safety threshold	P_ICP <= 20 mmHg (INV-1)	MODELLED
DI-006	Per-segment flow safety	F_i <= F_MAX (INV-2)	MODELLED
DI-007	Prediction lead time	>= 24h before symptoms	MODELLED
DI-008	Revision rate reduction	30%	MODELLED
DI-009	Number of segments	4 (V0 prototype)	MODELLED
DI-010	Biocompatibility (ISO 10993)	UNKNOWN	UNKNOWN
DI-011	Sterilization compatibility	UNKNOWN	UNKNOWN
DI-012	EMC	UNKNOWN	UNKNOWN

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	Multi-segment catheter geometry	ABSENT	Catheter diameter; Segment length; Segment count (production); Lumen
DO-002	Flow sensor specifications	CONCEPTUAL	
DO-003	Bayesian predictor algorithm	MODELLED	
DO-004	Alpha-distribution controller	MODELLED	

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
Number of segments N	MODELLED (4 in V0 prototype)	count	R332 decisive experiment	
Per-segment flow safety F _{inv-2}	MODELLED (INV-2)	mL/min	R332 mechanism	
ICP safety threshold	MODELLED (P_ICP <= 20 mmHg AND F _{inv-1} > 10 mL/min) — FAILURE	mmHg	R332 decisional invariance	R332 decisional invariance; graceful degradation claim replaces
Prediction lead time	MODELLED target 24h; achieved 48h (FAILS target)	hours	R332 modelled_only	
Revision rate reduction	MODELLED 30% target	%	R332 modelled_only	
Segment geometry (r, L)	UNKNOWN production dimensions	mm	Design choice	

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
Dual-invariant conflict	P_ICP > 20 mmHg AND F _{inv-2} < 10 mL/min	Graceful degradation to R332 simulation (2025 release)	R332 simulation (2025 release)	UNKNOWN whether graceful degradation suffices in v
Prediction lead time	insufficient fails to provide 24h headw	Improved sensor sensitivity	Hybrid obstruction simulation	UNKNOWN whether improved time with improved senso
Multiple simultaneous occlusions	Multiple segments occlude faster than control detection	Multi-segment obstruction	UNKNOWN	UNKNOWN multi-occlusion rate
Sensor failure	Per-segment flow sensor fails	Sensor redundancy + fault detection	Sensor failure simulation	UNKNOWN sensor failure rate
Catheter obstruction (Separate tissue block phenom)	Multi-segment redundancy	Obstruction tolerance test	UNKNOWN	UNKNOWN whether multi-segment reduces overall o
Manufacturing complexity	Multi-segment extrusion + sensor development	Process development	Qualification multi-segment	UNKNOWN achievable Cpk

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
Dual-invariant conflict (FALSIFIED)	P_ICP > 20 AND F_i > F_max simultaneous	Graceful degradation protocol	UNKNOWN if graceful degradation suffices in vivo
Prediction lead time insufficient	24h target not achieved (~14h)	Better sensors + algorithm refinement	UNKNOWN achievable lead time
Multiple simultaneous segment occlusions	Multi-segment redundancy exhausted	Increase N; rapid detection	UNKNOWN multi-occlusion rate
Sensor failure	Per-segment flow sensor fails	Sensor redundancy + fault detection	UNKNOWN sensor failure rate
Catheter obstruction	Debris/tissue blocks lumen	Multi-segment redundancy + obstruction detection	UNKNOWN whether multi-segment reduces overall rate
Manufacturing complexity	Multi-segment extrusion complex	Process development + vendor qualification	UNKNOWN achievable Cpk

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	Multi-segment peak ICP < single-segment peak ICP	Multi-segment + COTS sensors + scenarios	80% scenarios 30 obstruction not fails	NOT TESTED
V-002	Strict dual-invariant (P_ICP <= 20 AND F_i < F_max)	Computational simulation	Both invariants maintained	FALSIFIED (peak 22 > 20 mmHg)
V-003	Graceful degradation	Computational simulation	Multi-segment survival > single-segment	FALSIFIED (peak 22 vs 59 mmHg, survival)

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	30% revision rate reduction in clinical use	Clinical trial (IDE required)	>=30% reduction vs standard shunt	NOT PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
Catheter body	Silicone elastomer	External: FDA-cleared shunts use silicone (e.g. Medtronic, Becton Dickinson)	ISO 10993 (to 62009) + biocompatibility	ENGINEERING_CANDIDATE — not selected, not verified
Catheter body	Polyurethane	External: some CSF shunt catheters use polyurethane	ISO 10993 + mechanical testing	CANDIDATE — not selected, not verified

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	Multi-segment catheter body	1	CUSTOM_COMPONENT	UNKNOWN — likely silicone or polyurethane	UNKNOWN	CRITICAL	Biocompatibility test + biocompatibility
02	Per-segment flow sensor	4 (V0)	COTS_CANDIDATE	DATA (COTS)	Transonic (identified)	HIGH	Calibration + integration test
03	Microcontroller (Arduino Uno)	1	COTS_CANDIDATE	DATA (COTS)	Arduino (identified)	MEDIUM	Software validation + EMC

13. Manufacturing

Status: ENGINEERING_CANDIDATE — no process validated for multi-segment design

Candidate Processes:

- Silicone extrusion: Standard CSF catheter manufacturing method - Multi-segment design may require modified extrusion or assembly
- Injection molding (valve components): Standard valve manufacturing -

14. External Evidence

Source 1: Classification of the Cerebrospinal Fluid Shunt System

URL: <https://www.federalregister.gov/documents/2021/12/29/2021-28157/medical-devices-neurological-devices-classification-of-the-cerebrospinal-fluid-shunt-system>

Excerpt: Dec 29, 2021 — The Food and Drug Administration (FDA or we) is classifying the cerebrospinal fluid shunt system into class II (special controls).

Source 2: Miethke proGAV® Shunt - [accessdata.fda.gov](https://www.accessdata.fda.gov)

URL: https://www.accessdata.fda.gov/cdrh_docs/pdf6/K062009.pdf

Excerpt: Jan 12, 2007 — The Miethke proGAV® Programmable Shunt System is intended to shunt cerebrospinal fluid (CSF) from the lateral ventricles of the brain into the ... Read more

Source 3: FDA Warns of Potential Complications Between CSF

URL: <https://www.neurologyadvisor.com/news/fda-warns-of-potential-complications-between-csf-shunts-and-implantable-devices>

Excerpt: Jul 17, 2019 — The FDA warned close proximity between such devices and programmable CSF shunts may lead to unintended changes in valve settings and, in turn, ...

Source 4: HYDROCEPHALUS SHUNT SYSTEM, LOW/MEDIUM/

URL: <https://fda.innolitics.com/device/K895524>

Excerpt: Apr 10, 1990 — FDA 510(k) clearance ■ shunt is a device or combination of devices used to divert fluid from the brain ■ valved catheters, valves, ...

Source 5: Programmable CSF Shunt Valves: Radiographic Identification

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7965480>

Excerpt: by SS Lollis · 2010 · Cited by 75 — SUMMARY: The programmable CSF shunt valve has become an important tool in hydrocephalus treatment, particularly in the NPH population and in pediatric patients ... Read more

15. Transfer Boundary

YOU RECEIVE:

- + Multi-segment catheter concept + mechanism description
- + Bayesian predictor algorithm (225 ablation runs, rate-limiting confirmed)
- + Alpha-distribution controller with graceful degradation protocol
- + Dual-invariant falsification record (honest failure disclosure)
- + V0 bench prototype BOM (COTS sensors + Arduino)
- + Computational model (svMultiPhysics verified 1D within 16%)
- + This engineering dossier with external precedent

YOU MUST DEVELOP:

- > Production catheter design (dimensioned drawings, tolerances)
- > Implantable-grade flow sensor (COTS → implantable transition)
- > Production controller (Arduino → medical-grade MCU)
- > Manufacturing process for multi-segment catheter
- > Regulatory submission (PMA Class III)
- > Clinical validation evidence

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Achievable prediction lead time with production-grade sensors — UNKNOWN
- Multi-segment obstruction rate in vivo — UNKNOWN
- Sensor failure rate in chronic implant — UNKNOWN
- Multi-segment extrusion process capability (Cpk) — UNKNOWN
- Regulatory pathway (Class II 510(k) with substantial equivalence, or Class III PMA) — UNKNOWN
- Clinical benefit magnitude (30% revision reduction is MODEL_DERIVED, not validated) — UNKNOWN until trial
- Cost vs single-segment catheter — UNKNOWN (likely higher cost; needs value analysis)

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	V0 bench prototype (4-segment catheter, error)	Bayesian predictor algorithm	Prediction lead time	Achieve data 24h lead time (targeted for ready report)	80% of scenarios	
WP-02	Bayesian predictor algorithm	Controller with graceful degradation	Multi-segment extrusion process capability (Cpk)	1.33 for critical dimensions	18 weeks	
WP-03	Controller with graceful degradation	Multi-segment extrusion process	Biocompatibility specification	ISO 10993 report	12 weeks (external lab)	
WP-04	Multi-segment extrusion process	Biocompatibility specification	Biocompatibility specification	ISO 10993 report	12 weeks (external lab)	
WP-05	Biocompatibility specification	Biocompatibility specification	Biocompatibility specification	ISO 10993 report	12 weeks (external lab)	

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-06	Implantable-grade sensor (COSS template)	SensQ COSS template	Stability + drift	Sensor is chronic	Sensors required	10 weeks

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 4 governing equations and a 6-step build plan. The next experiment is clearly specified. Target buyer: Neuroshunt manufacturer (Medtronic, Integra, Miethke).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: Multi-segment shows no advantage over single-segment in $\geq 80\%$ of computational scenarios

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: Multi-segment shows no advantage over single-segment in $\geq 80\%$ of computational scenarios

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.